

SE - Overview of IT Industry

Session 3: Internet & Networking Basics
Internet Fundamentals: How computers communicate. Protocols: Define HTTP, HTTPS, FTP, SMTP (overview only, 5 mins each). Domain & DNS: Explain how domain names map to IP addresses. Demonstrate using nslookup google.com. Explain DNS hierarchy: root, TLD, domain, subdomain. IP Addressing: What is an IP (IPv4 vs IPv6)? HTTP Lifecycle: Request ? Response ? Status Codes (focus on 200, 404, 500). Trainer Demo: Use browser DevTools ? Network tab

Task :

Question :- 1

Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans :-

HTTP :- Hyper Text Transfer Protocol

HTTPS :- Hyper Text Transfer Protocol Secure

FTP :- File Transfer Protocol

SMTP :- Simple Mail Transfer Protocol

Question :- 2

What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans :-

- IPv4 :- IPv4 uses 32-bit address and look like 192.168.1.1
- IPv6 :- IPv6 uses 128-bit address and look like 2001:db8::1

Question :- 3

Run nslookup google.com on your system and write down the IP address you get.

Ans :-

Address :- 10.56.165.148

Question :- 4

What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans :-

Code	Meaning
200	OK – Request successful
301	Moved Permanently – page has a new URL
404	Not Found – Page does not exist
500	Internal Server Error – Server problem
403	Forbidden – Access denied

Question :- 5

Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans :-

Part	Type
.(root)	Root Domin
Com	TLD (Top Level Domin)
Google	Domain Name
www	Subdomain